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# Spectral Assignments and Reference Data

## $^1$H and $^{13}$C chemical shifts for bis(benzopyridinium) dibromides with semirigid aromatic linkers

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**The $^1$H and $^{13}$C NMR resonances for four bisquinolinium and four bisisoquinolinium dibromides were completely and unequivocally assigned, using the concerted application of one- and two-dimensional NMR techniques including nuclear Overhauser effect difference, DEPT, COSY, HETCOR and long-range $^{13}$C—$^1$H correlation spectroscopy. Atomic charges of the heterocyclic moieties of model cations such as 1-methylquinolinium and 2-methylisoquinolinium are shown to be linearly related to the $^{13}$C chemical shifts of the cationic heads of the title compounds. Copyright © 2002 John Wiley & Sons, Ltd.**

**KEYWORDS:** NMR; $^1$H NMR; $^{13}$C NMR; 2D NMR; bisquinolinium dibromides; bisisoquinolinium dibromides; AM1 Hamiltonian

## INTRODUCTION

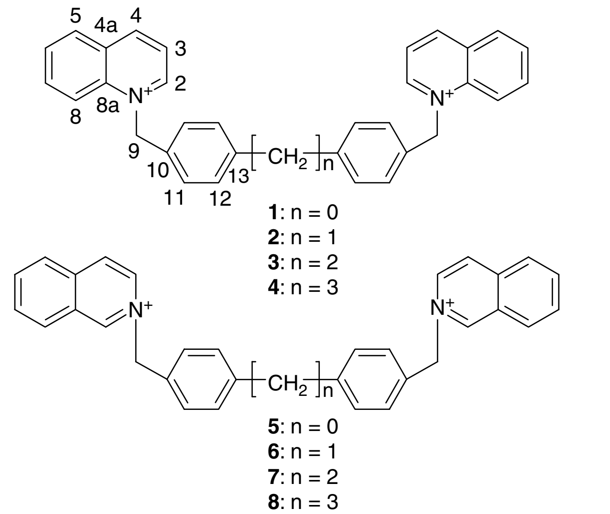
The considerable importance of quinolinium[1,2] and isoquinolinium[3] compounds in clinical medicine and their recent development as anticancer drugs have increased the number of studies of such compounds. These structures are of considerable medicinal, chemical and pharmacological interest.[4,5]

Recently, we described some bisquinolinium and bisisoquinolinium salts as anticancer drugs via the inhibition of choline kinase.[6] Although the structures of these compounds had been established by standard spectroscopic analysis [$^1$H NMR ($CD_3OD$), MS], we did not carry out a detailed NMR examination. In view of their biological importance we have now carried out a more detailed NMR study to gain more insight into their molecular structure prior to conducting further structure–activity investigations. This paper reports the unequivocal assignments for the $^1$H and $^{13}$C resonances of the title compounds **1–8** (Scheme 1), using one- and two-dimensional NMR techniques. Finally, correlations between the atomic charges (ACs) of model 1-methylquinolinium and 2-methylisoquinolinium cations and the $^{13}$C chemical shifts of the title compounds were obtained.

## EXPERIMENTAL

Compounds **1–8** were prepared and purified as described previously.[6] The $^1$H and $^{13}$C NMR spectra were recorded on a Bruker AM-300 spectrometer operating at 75.479 MHz for $^{13}$C and 300.160 for $^1$H in $CD_3OD$ and on a Bruker ARX 400 spectrometer operating at 400.132 MHz for $^1$H and 100.623 MHz for $^{13}$C (a concentration of ca 27 mg ml$^{-1}$ in all cases). The centre of each peak of $CD_3OD$ [3.31 ppm ($^1$H) and 49 ppm ($^{13}$C)] was used as an internal reference in a 5 mm $^{13}$C/$^1$H dual probe (Wilmad, No. 528-PP). The temperature of the sample was maintained at 297 K. The accuracy of the chemical shifts was better than ±0.01 ppm.

One-dimensional NOE difference experiments on **3** and **7** were performed by irradiation for 4 s in series of eight scans with alternating on- and off-resonance. The irradiating power was set low to achieve selectivity.




**Scheme 1.** Structures and numbering of the compounds studied.

The following parameters were used for the DEPT experiments: PW (135°), 9.0 μs; recycle time, 1 s; $1/2J(CH) = 4$ ms; 65 536 data points acquired and transformed from 1024 scans; spectral width, 15 kHz; and line broadening, 1.3 Hz.

The $^1$H–$^1$H chemical shift correlation (COSY) utilized the sequence described by Nagayama *et al.*[7] with a mixing pulse of 45°; 16 scans were measured for 128 values of $t_1$ to give a total data matrix of 128 × 2048 complex points with a sweep width of 2.7 kHz.

The $^1$H–$^{13}$C chemical shift correlation (HETCOR) utilized the sequence described by Rutar[8] and Wilde and Bolton[9] with delays optimized for $J$(CH) values of 170 Hz.; 32 scans were measured for 128 values of $t_1$ to give a total data matrix of 128 × 4096 complex points. The carbon $F_2$ spectral width was 15 kHz and the proton $F_1$ spectral width was 3.2 kHz.

The long-range $^1$H–$^{13}$C heteronuclear multiple bond connectivity (HMBC) was recorded on a Bruker ARX 400 spectrometer and utilized the sequence described by Bax and Morris[10] with delay values optimized for $^3J$(CH) values of 10 Hz; 16 scans were measured for 512 values of $t_1$ to give a total data matrix of 512 × 8192 complex points. The carbon $F_2$ spectral width was 20 kHz and the proton spectral width was 4.2 Hz.

Atomic charges for 1-methylquinolinium and 2-methylisoquinolinium cations were calculated on a Silicon Graphics IRIS 4D workstation, running SYBYL 6.5[11] at the semiempirical level, using the AM1[12] Hamiltonian of MOPAC (Version 6.0).[13]

## RESULTS AND DISCUSSION

The results are given in Tables 1 and 2. Remarkably, none of the nine aromatic protons overlaps at 300 MHz in $CD_3OD$ where the total shift scale is only 2.4 ppm for **1–4** and 2.8 ppm for **5–8**.

The $^1$H NMR spectrum of **3** started by the NOE enhancement of the doublets at 9.53 and 8.51 ppm, in addition to H-11 of the benzylic spacer at 7.22 ppm upon irradiation of the H-9 singlet at 6.32 ppm. The $\alpha$-proton of pyridinium ions[14] shows the largest downfield shift, and accordingly the 9.53 ppm peak was assigned to H-2. In the aromatic region, H-8 can be easily differentiated from H-11 on the basis of their chemical shifts. The former appears at lower field (at 8.51 ppm) owing to the deshielding effect of the pyridinium nucleus. The H-2, H-8 and H-11 signals serve as entry points for spectral interpretation by means of the $^1$H–$^1$H COSY experiment. The $^{13}$C resonances were obtained by the data provided by DEPT and $^{13}$C–$^1$H HETCOR experiments. Short-range correlations provided an unambiguous assignment of the methylene carbon and all the methine carbons. The long-range correlations determined by HMBC

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**Table 1.** $^{1}H$ NMR chemical shifts (ppm), multiplicities and $J$(H,H) coupling constants (Hz, in italics) in $CD_3OD$ for the symmetrical (1,1′-disubstituted bisquinolinium) dibromides **1–4**[a] and (2,2′-disubstituted bisisoquinolinium) dibromides **5–8**[a] with aralkyl linking groups

| Hydrogen | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|
| 1 | — | — | — | — | 10.16 s | 10.09 s | 10.09 s | 10.11 s |
| 2 | 9.60 dd | 9.53 dd | 9.53 d | 9.51 dd | — | — | — | — |
| | *5.9, 1.4* | *5.8, 1.4* | *5.9* | *5.8, 1.1* | | | | |
| 3 | 8.20 m | 8.18 m | 8.16 dd | 8.18 m | 8.72 dd | 8.65 dd | 8.66 dd | 8.67 dd |
| | | | *8.3, 1.4* | | *6.7, 1.5* | *6.8, 1.5* | *6.9, 1.5* | *6.8, 1.2* |
| 4 | 9.31 d | 9.28 d | 9.29 d | 9.28 d | 8.49 d | 8.44 d | 8.48 d | 8.46 d |
| | *8.3* | *8.3* | *8.3* | *8.3* | *6.7* | *6.8* | *6.9* | *6.5* |
| 5 | 8.46 dd | 8.44 dd | 8.45 d | 8.44 d | 8.30 d | 8.27 d | 8.30 d | 8.28 d |
| | *8.0, 1.0* | *8.2, 1.4* | *8.2* | *8.2* | *8.2* | *7.7* | *8.0* | *8.2* |
| 6 | 8.01 dd | 8.00 ddd | 8.02 dd | 8.01 dd | 8.24 ddd | 8.22 ddd | 8.25 dd | 8.23 ddd |
| | *8.0, 0.7* | *8.2, 7.4, 0.7* | *7.0, 0.9* | *7.5, 6.2* | *8.2, 6.7, 1.0* | *7.7, 6.7, 1.0* | *6.6, 0.9* | *8.2, 7.0, 0.8* |
| 7 | 8.20 m | 8.18 m | 8.23 dd | 8.18 m | 8.07 ddd | 8.04 ddd | 8.07 dd | 8.05 ddd |
| | | | *8.3, 1.4* | | *8.4, 6.7, 1.2* | *8.3, 6.7, 1.5* | *6.6, 1.2* | *8.3, 7.0, 0.9* |
| 8 | 8.53 d | 8.50 d | 8.51 d | 8.52 d | 8.54 d | 8.50 dd | 8.52 dd | 8.51 d |
| | *8.6* | *8.6* | *9.0* | *9.0* | *8.4* | *8.3, 1.0* | *8.3, 0.9* | *8.3* |
| 9 | 6.41 s | 6.30 s | 6.32 s | 6.30 s | 6.05 s | 5.94 s | 5.94 s | 6.10 s |
| 11 | 7.45 d | 7.23 d | 7.22 d | 7.22 d | 7.68 d | 7.30 d | 7.29 d | 7.27 d |
| | *8.3* | *8.3* | *8.2* | *8.1* | *8.5* | *8.2* | *8.1* | *8.0* |
| 12 | 7.66 d | 7.29 d | 7.27 d | 7.28 d | 7.73 d | 7.50 d | 7.48 d | 7.50 d |
| | *8.3* | *8.3* | *8.2* | *8.1* | *8.5* | *8.2* | *8.1* | *8.0* |
| —$CH_2$— | — | 3.95 s | | | | 3.99 s | | |
| —$CH_2$—C— | | | 2.87 s | | | | 2.93 s | |
| —$CH_2$—C—C— | | | | 2.63 t | | | | 2.63 t |
| | | | | *7.7* | | | | *7.7* |
| —C—$CH_2$—C— | | | | 1.88 q | | | | 1.88 q |
| | | | | *7.7* | | | | *7.7* |

[a] In ppm from the centre of $CD_3OD$ (3.31 ppm).

**Table 2.** $^{13}C$ chemical shifts (ppm) in $CD_3OD$ for the symmetrical (1,1′-disubstituted bisquinolinium) dibromides **1–4**[a] and (2,2′-disubstituted bisisoquinolinium) dibromides **5–8**[a] with aralkyl linking groups

| Carbon | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|
| 1 | — | — | — | — | 151.14 | 150.98 | 150.97 | 150.92 |
| 2 | 151.04 | 150.84 | 150.81 | 150.84 | — | — | — | — |
| 3 | 123.29 | 123.23 | 123.24 | 123.21 | 135.61 | 135.53 | 135.51 | 135.53 |
| 4 | 149.82 | 149.68 | 149.67 | 149.67 | 127.85 | 127.79 | 127.79 | 127.78 |
| 4a | 131.94 | 131.89 | 131.90 | 131.84 | 139.16 | 139.10 | 139.14 | 139.12 |
| 5 | 132.21 | 132.16 | 132.17 | 132.16 | 128.61 | 128.59 | 128.62 | 128.58 |
| 6 | 131.49 | 131.43 | 131.45 | 131.47 | 138.66 | 138.60 | 138.64 | 138.59 |
| 7 | 137.36 | 137.27 | 137.24 | 137.27 | 132.78 | 132.75 | 132.81 | 132.75 |
| 8 | 120.32 | 120.29 | 120.36 | 120.29 | 131.80 | 131.75 | 131.74 | 131.73 |
| 8a | 139.68 | 139.63 | 139.66 | 139.63 | 129.28 | 129.23 | 129.27 | 129.25 |
| 9 | 61.79 | 61.82 | 61.96 | 61.95 | 65.15 | 65.27 | 65.34 | 65.41 |
| 10 | 134.05 | 132.33 | 132.04 | 132.33 | 134.43 | 132.81 | 132.47 | 132.30 |
| 11 | 129.26 | 131.01 | 130.73 | 130.62 | 130.91 | 131.13 | 130.84 | 130.71 |
| 12 | 129.06 | 128.95 | 128.73 | 128.80 | 129.20 | 130.50 | 130.25 | 130.30 |
| 13 | 142.10 | 143.68 | 144.28 | 144.68 | 142.68 | 144.22 | 144.87 | 145.57 |
| —$CH_2$— | | 41.85 | | | | 42.03 | | |
| —$CH_2$—C— | | | 38.15 | | | | 38.22 | |
| —$CH_2$—C—C— | | | | 35.94 | | | | 35.99 |
| —C—$CH_2$—C— | | | | 33.90 | | | | 33.92 |

[a] In ppm from the centre of $CD_3OD$ (49 ppm).

# Spectral Assignments and Reference Data

**Table 3.** Average chemical shifts, standard deviations (2,2′-disubstituted for the quinolinium and isoquinolinium ring carbon atoms of (1,1′-disubstituted bisquinolinium) and bisisoquinolinium) dibromides in $CD_3OD$ and atomic charges for model compounds

| Atom | $\delta^{13}C(CD_3OD)$ for the quinolinium moiety | Atomic charge[a] | $\delta^{13}C(CD_3OD)$ for the isoquinolinium moiety | Atomic charge[a] |
|---|---|---|---|---|
| N-1 | — | −0.041 | — | — |
| N-2 | — | — | — | −0.036 |
| C-1 | — | — | 151.00 ± 0.08 | 0.059 |
| C-2 | 150.88 ± 0.09 | 0.054 | — | — |
| C-3 | 123.24 ± 0.03 | −0.186 | 135.54 ± 0.04 | −0.073 |
| C-4 | 149.71 ± 0.06 | 0.128 | 127.80 ± 0.03 | −0.087 |
| C-4a | 131.89 ± 0.04 | −0.077 | 139.13 ± 0.02 | 0.019 |
| C-5 | 132.18 ± 0.02 | −0.076 | 128.60 ± 0.02 | −0.124 |
| C-6 | 131.46 ± 0.02 | −0.103 | 138.62 ± 0.03 | −0.048 |
| C-7 | 137.28 ± 0.04 | −0.039 | 132.77 ± 0.02 | −0.095 |
| C-8 | 120.32 ± 0.03 | −0.157 | 131.76 ± 0.03 | −0.086 |
| C-8a | 139.65 ± 0.02 | 0.044 | 129.26 ± 0.02 | −0.093 |

[a] Calculated using the AM1 method[12] to derive MOPAC charges.[13]

were useful in the assignment of quaternary carbons C-4a, -8a, -10 and -13, which showed similar chemical shifts. Analysis of the spectra of **1, 2** and **4** was then easily derived. It should be pointed out that C-13 always appears downfield compared with C-10 in **1–4**. The same holds for the bisisoquinolinium salts **5–8** (see Table 2).

Although $^{1}H$ and $^{13}C$ NMR spectra of 1-ethylquinolinium iodide (in DMSO-$d_6$) have been reported previously,[15] two of the proposed $^{13}C$ shifts were found to be interchanged. The values of C-5 and C-8, which were reported as being 118.59 and 130.30 ppm, respectively, are actually swapped, as can be deduced from the unequivocal assignment of compounds **1–4**. The $^{1}H$ and $^{13}C$ NMR chemical shifts of the bisisoquinolinium salt **7** were established following the similar study carried out on **4**. Finally, comparison of chemical shifts of **5, 6** and **8** with respect to those of **7** allowed their unequivocal assignment.

Good correlations were obtained between atomic charges (ACs) of the 1-methylquinolinium and 2-methylisoquinolinium cations and the $^{13}C$ chemical shifts (CSs) for **1–4** and **5–8** (Table 3), respectively, using the AM1[12] method to derive MOPAC charges:[13]
for **1–4**:

$$AC = -1.31\ (\pm 0.16) + 0.01\ (\pm 0.00)\ \ CS$$
$$n = 9,\ r = 0.951,\ s = 0.034$$

for **5–8**:

$$AC = -1.08\ (\pm 0.16) + 0.01\ (\pm 0.00)\ \ CS$$
$$n = 9,\ r = 0.923,\ s = 0.024$$

where $n$ is the number of carbon atoms, $r$ is the correlation coefficient, $s$ is the standard deviation and data in parentheses are 95% confidence intervals.

In conclusion, the calculations carried out confirm the good correlations between ACs of the carbon atoms of the nitrogen heterocycles using the AM1 method. The set of chemical shifts presented here can serve as a reference for the unsubstituted parent compounds **1–8** for the application of the substituent additivity effects on more complicated benzopyridinium salts, as a tool for structural characterization.

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